

ULPF — Universal Log Pre-processing Framework

Overview

Modern enterprises generate logs from diverse sources such as **firewalls, IDS/IPS, proxies, web servers, network monitors, and syslog systems**. These logs differ in format, structure, and vendor-specific fields, making unified analysis and correlation difficult.

ULPF provides a universal, lossless preprocessing layer that converts heterogeneous logs into a common **ECS 8.11-based normalized format**, while preserving the original event for traceability.

Architectural Approach

ULPF follows a **modular, plugin-based architecture**:

- **ML-based classification** identifies the probable log type.
- A **plugin registry** dynamically routes events to the appropriate parser.
- **Source-specific plugins** parse and map fields into the common schema.
- The **normalization layer** produces standardized ECS-based events.
- **Raw events and metadata are preserved** throughout processing.
- **Low-confidence, unsupported, or failed events are quarantined** rather than silently discarded.

Design Goals

Universal — Support heterogeneous log formats through a common processing architecture.

Extensible — New log sources can be onboarded through plugins without modifying the core pipeline.

Lossless — Preserve original events and processing metadata for complete traceability.

Deployment Ready — Self-contained and suitable for containerized, air-gapped environments.

Current Prototype

8 parser plugins covering Cisco ASA, Snort, Syslog, Zeek, Proxy, and Web Access logs, with ML-based routing and quarantine handling.

